

SOURCE™

Introduction

The SOURCE™ Hybrid 6kW System by AMEC Technology, is an all-in-one solar + storage solution for residential and light commercial use, combining a high-efficiency hybrid inverter, advanced MPPT controller, intelligent BMS, and AI-powered load optimization in a compact, wall-mounted liquid-cooled design. Delivering 6 kW continuous AC power, supporting up to 7 kW solar input, and offering 6.5 kWh LFP battery storage, it enables seamless on/off-grid operation with smart forecasting, time-of-use optimization, and real-time monitoring via the SOURCE™ app—reducing grid reliance and ensuring reliable backup during outages.



System Overview

Model	SOURCE™ Hybrid 6 kW Liquid-Cooled System
Form factor	Wall-mounted design with integrated liquid cooling & a hybrid aluminum-plastic enclosure
Cooling	Active liquid-cooled with smart thermal management
Smart power integration	Intellisync - AI based Adaptive Power Management system (Smart Load Shifting, Forecasting)
Mobile application	Real-time + Historical Power Data [BLE 5.2, Wi-Fi, Cloud Dashboard]
User interface	Touch Display + App Control (iOS & Android)
Modes	Backup, Time-of-Use, Off-Grid (Islanding), Solar Priority

Inverter Module

Continuous power output	6 kW
AC output voltage	230V AC $\pm 5\%$, 50Hz
DC input range	42V – 60V
Grid capabilities	On-grid / Off-grid Compatible
Surge Power/ protection (10s)	9 kW
Conversion efficiency	$\geq 98\%$
Switch time	Under 20 ms
Protection System	Anti-islanding, Short Resistance, Grid Synchronization

Battery Module

Battery capacity	6.5 kWh
Nominal voltage	51.2 V
Chemistry	LiFePO4 (LFP)
Warranty	10 Years
Operational temperature	-10°C to 55°C
Protection	Integrated BMS (Over/Under Voltage, Temp, Short Circuit)

MPPT

Max PV input power	7 kW
Max output from MPPT	6.6 kW
Solar efficiency	≥ 99% MPPT Tracking Efficiency
Protection	Integrated ACDB & DCDB (Breaker, SPD & Fuses)

PV Solar panel

Panel type	TopCon N-Type Bifacial Monocrystalline PERC
Rated power	575W per panel
Efficiency	≥ 22.2%
Number of Panels	12 panels (Total: 7 kW)
Frame	Pure Galvanized steel Structure
Open circuit voltage (VOC)	96V total (approx.)
Warranty	25 Years Performance Warranty

Auto PV panel cleaning system (Touchless Clean-Tech)

Cleaning method	Soft Brush with Anti-static Roller + water Spray
Automation	Fully Programmable via Mobile App or AI-driven Algorithms
Cleaning frequency	AI Adaptive or Scheduled (Daily/Weekly)
Motor type	12V DC Brushless Motor (IP65 Rated)
Control interface	Via SOURCE™ App (BLE 5.2 / Wi-Fi)
System warranty	2 Years Standard

Mechanical

Housing/casing	Die Cast Aluminum Alloy
Front panel	ABS + PC Blend Plastic Enclosure (UV & Impact Resistant)
Thermal management system	Active Liquid Cooling with Thermal Reservoir
Total weight	65 kg
Dimension (L*W*D)	930 mm*485 mm*200 mm

Package Includes

1x SOURCE™ 6.5 kW Hybrid Power Unit
1x ACDB & DCDB Box with Breakers, SPD & Fuse
1x Mounting Bracket Kit and pipe wire fitting
12x PV PANNEL UNIT 6.9Kw with Galvanized steel frame structure
User Manual + QR for App Installation

Terms

- All values are measured at 25 °C (77 °F) at the beginning of product life under standard test conditions, unless otherwise specified.
- Continuous power and peak ratings are subject to installation quality, cable sizing, and breaker ratings as per electrical standards.
- For maximum performance, system must be installed with an appropriately rated AC/DC breaker and conductors sized according to local regulations.
- Performance data is based on typical solar-to-battery and battery-to-load use cases; actual performance may vary depending on site conditions and load profile.
- Reliable internet connectivity is required for firmware updates, remote diagnostics, and app-based monitoring; cellular connectivity is recommended only as a backup and is subject to network coverage.
- Performance may be de-rated at operating temperatures above 40 °C (104 °F) or below 0 °C (32 °F).
- Physical dimensions and weight include all covers, mounting hardware, and protective enclosures.
- Specifications are subject to change without prior notice for product improvements.